Can Personalized Advice Improve Investor Behavior? Evidence from a Multi-Agent Simulation of Retail Investors

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Abstract

Retail investors are known to exhibit behavioral biases such as the disposition effect, and personalized advice could help mitigate them. Yet it is hard to know whether such advice works: testing it on real investors at scale is difficult, and observational data cannot show how the same investor would have traded under different advice from the same initial state and under the same market conditions. Recent advances in large language models (LLMs) make it possible to imitate investors with distinct tendencies, evolving beliefs, and social interaction, so we build a multi-agent simulation of 100 heterogeneous retail investors trading Samsung Electronics under historical prices and news, whose aggregate trading broadly tracks observed Korean retail investor flows. After a common trading period, we copy each investor's state and compare no advice, general advice, and personalized advice grounded in the investor's own trading history, examining both executed trades and stated rationales. In this exploratory run, personalized advice was associated with a weaker disposition effect than general advice, reflecting a larger increase in loss-state selling than in gain-state selling compared with no advice. Responses varied across investors, and citing the advice in stated reasoning did not guarantee improved trading. Because the two advice texts also differ in authorship, length, and specificity, the comparison concerns advice designs rather than personalization alone. By comparing the same investor under different advice from an identical starting state and market path, while observing both executed trades and stated rationales, our simulation testbed makes it possible to evaluate which trades advice changes for whom and where it fails questions that observational data cannot answer.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**.

Keywords

Multi-Agent Simulation, LLM agents, Behavioral finance, Disposition effect, Financial advice, Robo-advisor

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1 Introduction

Recent advances in large language models (LLMs) have broadened their applications in finance, including financial text analysis, portfolio construction, and investment advice [10, 16, 25]. LLM-based agents combine these capabilities with information retrieval, memory, and reasoning to perform financial tasks over successive decisions [4, 12, 27]. These agents also offer a means of representing investors whose decisions reflect different preferences and experiences. Multi-agent simulations model investor populations to examine how individual decisions and social interactions shape collective trading behavior [26, 28].

Understanding retail investor behavior requires detailed trading records, but access to individual account data remains limited [9, 15]. Transactions alone also reveal little about how investors interpret information or revise their beliefs, and observational data cannot show how the same investor would respond to different advice under identical market conditions. Simulation offers a complementary route by making investor states observable and allowing alternative conditions to be compared, provided that simulated trading reflects patterns observed among actual investors.

We introduce a multi-agent simulation of retail investors in the Korean equity market, combining heterogeneous investment tendencies, evolving beliefs, and social interactions under historical prices and news. Agents act as price takers, preserving a common market path across experimental conditions. As a check on the environment, simulated daily net purchases reproduce much of the observed direction of retail investor flows and agree more than simple directional baselines.

Retail investors show the disposition effect, averaging down, and buying after price surges [1, 15, 22], and the benefits of advice vary with their prior behavior [5, 19]. General principles are easy to state but hard to apply to one's own trading, whereas advice grounded in an investor's own trading history may connect them

to recurring behavior; we examine this as a hypothesis. After a common trading period, we copy each investor's state and compare, under identical prices and news, no advice, general advice based on common investment principles, and personalized advice written from each agent's profile, beliefs, and trading history. We ask (Q1) how measured trading patterns change when advice is given, and (Q2) how the size and distribution of these changes differ between general and personalized advice.

In this exploratory run, personalized advice was associated with a weaker disposition effect than general advice, driven by a larger increase in loss-state selling. Responses varied across investors, and citing the advice in stated reasoning did not guarantee a change in trading. Because the two advice designs differ in authorship, length, and specificity, the comparison concerns advice designs rather than personalization alone.

The contributions of this study are as follows.

- We develop a multi-agent simulation of heterogeneous retail investors that supports matched comparisons of advice conditions, and assess its aggregate alignment with observed trading in the Korean equity market.
- We quantify three behavioral patterns from individual executed trades, extending the analysis of simulated investors beyond aggregate market outcomes.
- We compare general and personalized advice against no advice to examine which trades change, which investors general advice leaves largely unchanged, and whether stated reasoning matches executed trades.

2 Related Work

2.1 LLM-Based Agent Simulation

LLM-based agent simulations combine language generation with memory and environmental feedback to model behavior over repeated interactions. Generative Agents integrates memory retrieval, reflection, and planning to simulate individual routines and social interactions [17]. AgentSociety couples agents with environmental feedback and parallelized interaction mechanisms to support large populations [29]. Persistent memory and interaction with the environment allow simulated behavior to evolve as agents accumulate experience and encounter new information.

Economic and financial simulations use these architectures to study decisions under heterogeneous preferences and information. EconAgent models household labor and consumption decisions and examines the resulting macroeconomic patterns [13]. StockAgent evaluates trading responses to changes in market information and external events [28]. TwinMarket combines investor beliefs with social interaction to study collective trading dynamics [26]. Lopez-Lira uses heterogeneous LLM traders in an order-driven market to study strategy adherence, price discovery, and liquidity provision [14]. Our study complements this literature by comparing simulated retail net purchases with observed trading under historical prices and news. Fixed prices and news allow comparisons of advice conditions along the same historical market path.

2.2 Investor Behavior and Financial Advice

Empirical studies document systematic differences in how retail investors respond to gains, losses, and salient information. The disposition effect describes a preference for realizing gains over losses, supported by evidence from brokerage records [15, 20]. Individual investors also tend to buy stocks that attract attention through news, unusual volume, or extreme returns [1], and to add to positions held at a loss more often than those held at a gain [22]. Evidence from Korean accounts links gain realization to foreign investor inflows [18] and identifies heterogeneous trading patterns with different profit distributions [7]. These findings motivate evaluating advice against several behavioral patterns while examining variation in responses across investment tendencies.

The effects of financial advice depend on investor uptake and content. Offering unbiased advice free of charge has produced low acceptance and adherence [3], while human advisors exert substantial influence over client allocations with limited customization [6]. Robo-advisor adoption has been associated with reductions in the disposition effect and trend chasing, but does not eliminate these patterns [5]. Diversification and welfare benefits also depend on investors' portfolios before adoption [19]. Conversational LLM advice can improve investment choices when investor preferences are elicited accurately, but unsuccessful elicitation can lead to unsuitable recommendations [23]. LLM investment analysis itself can exhibit systematic biases [11], and recommended portfolios can display concentration and trend chasing risks [24]. Our study compares advice grounded in individual trading histories with general guidance and no advice, holding common initial investor states and subsequent market conditions fixed.

3 Simulation and Advice Design

We model 100 retail investors as interacting LLM agents that interpret information, trade, and exchange views under controlled market conditions.

3.1 Overview and Decision Cycle

Each agent trades twice daily: at 09:00 at the actual opening price (AM) and at 15:30 at the closing price (PM). Orders are filled in full and do not change prices.

Figure 1(a) summarizes one trading day. To make a trade, an agent first forms its current interpretation of the news and community posts it has read through its profile, then assesses outlook and risk using both beliefs (Section 3.2), prices, market indicators, and its portfolio. It selects buy or sell and a quantity, recording a rationale and risk-management statement. All agents decide before their orders execute together; responses in an invalid format are requested again. The resulting trades then inform the next belief update. After the PM trade and belief update, agents interact in the community, and the posts they receive enter the next AM decision (Section 3.3). When advice is assigned, it enters analysis, trading decisions, and belief updates (Section 3.4).

At each AM or PM decision, agents receive information available at that time Table 1. Prompts exclude future prices, future news, and actual net purchases by investor category. Feedback on earlier trades is supplied only after the relevant observation period has elapsed: the next decision, one trading day, or five trading days. Alibaba does not publish an exact knowledge cutoff for the model, but the dated snapshot we use Table 4 was released on February

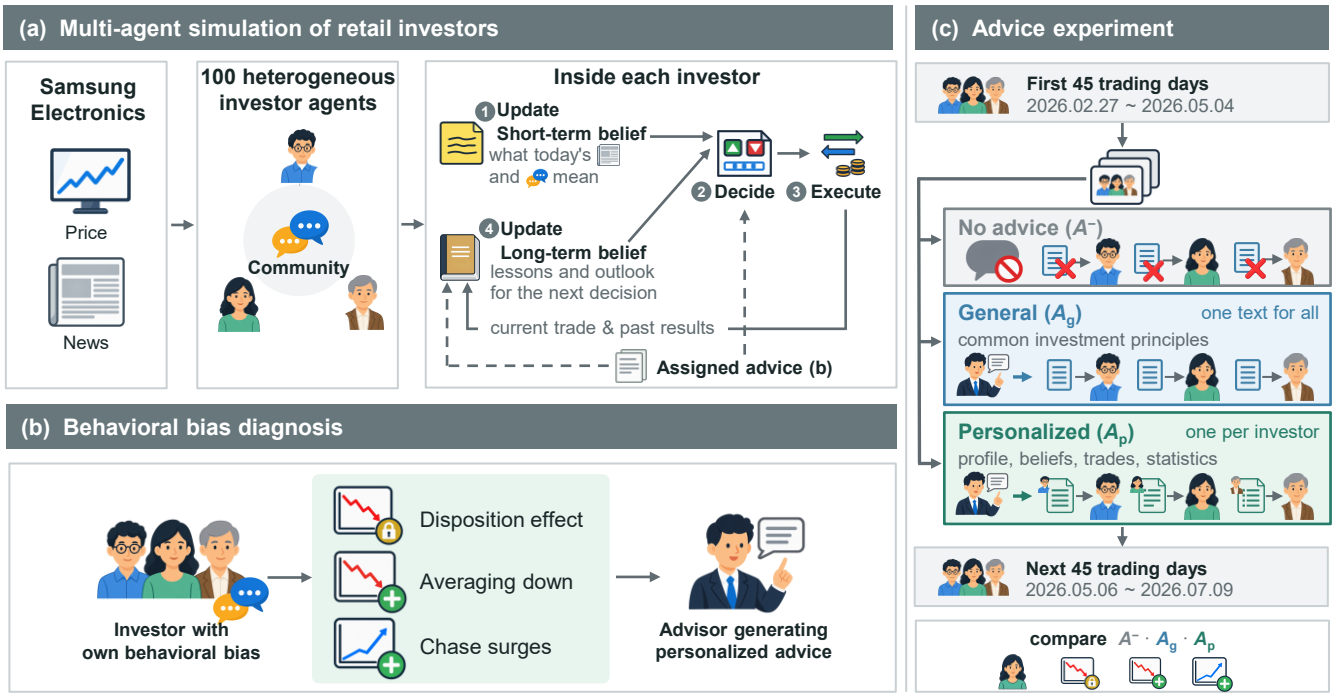


Figure 1: Overview of the study. (a) LLM investor agents trade Samsung Electronics under historical prices and news. (b) Behavioral biases measured from executed trades inform personalized advice. (c) After a shared 45-day period, each agent's state is copied into no-advice (A^-), general (A_g), and personalized (A_p) branches and compared over 45 more days. Icons are schematic.

Table 1: Information available at each decision event. Access to community information depends on news depth (D0–D2; Section 3.2).

Information	AM (09:00)	PM (15:30)
News	Articles from previous trading day 15:30 to current day 08:59	Articles from current day 08:59 to 15:30
Prices	Current open (execution price), previous close, open-to-previous-close return	Current close (execution price), open-to-close return
Market indicators	Return, volume change, 5- and 20-day moving averages, 20-day volatility	Same indicator types
Community	Previous day's top five posts; D1/D2 agents also receive the posts they selected	None before the decision
Portfolio	Cash, shares, average acquisition cost, unrealized profit/loss, feasible quantities	Same fields
Beliefs	Previous long-term belief and current short-term belief	Same structure

23, 2026, before the simulation period begins, so its training data cannot include the prices and news observed during the simulation.

3.2 Investor Personas and Beliefs

An investor profile, or *persona*, is a written description of an agent's investment tendencies. We select 100 of the 1,000 synthetic profiles released with TwinMarket [26], considering their behavioral attributes, and translate them into Korean. We index the 100 agents as A001–A100 and refer to individual agents by these identifiers. We add entry timing, news access, and initial cash; each attribute is expressed as a fixed sentence in the agent's instructions.

Selling-bias tendency describes how readily an agent sells after gains or continues buying after losses. Lottery preference describes its attraction to risky stocks offering large potential gains. Entry timing, assigned randomly, specifies whether new purchases favor falling prices, rising prices, or neither.

News access has three levels: article titles only (D0), titles and summaries (D1), and titles and summaries plus searches for up to five articles from the preceding seven days (D2). It is assigned

randomly, except that the nine influencers receive only D1 or D2 because D0 agents cannot post (Section 3.3). These levels also determine community access. Table 2 reports the profile composition. Appendix B provides attribute-level instruction summaries Table 10 and shows how attributes co-occur Tables 11 and 12. Diversification and past performance describe the assigned profiles, not additional holdings or realized returns in this experiment. Because profiles explicitly assign tendencies, observed biases do not show that an unconditioned LLM develops them spontaneously. Behavioral scores are calculated from actual trades.

Table 2: Persona attributes and cohort composition ($N = 100$). Slash-separated counts follow the order of the listed levels. Attributes marked † are added in this study; the others follow TwinMarket.

Attribute	Levels	Counts
User type	Ordinary / small influencer / large influencer	91/8/1
Selling-bias tendency	High / medium / low	33/34/33
Lottery preference	High / medium / low	30/6/64
Past performance	High / medium / low	30/36/34
Diversification	Diversified / concentrated	68/32
Analysis strategy	Technical / fundamental	60/40
Entry timing†	Contrarian / neutral / momentum	50/35/15
News depth†	D0 / D1 / D2	30/55/15
Initial cash†	KRW 100 million / 1 billion	73/27

An agent's profile stays fixed, while its views change as information arrives. We represent these views through two six-dimensional text representations: a short-term belief (STB) and a long-term belief (LTB). The short-term belief (STB) interprets current news and community posts through the profile. The long-term belief (LTB) combines its previous version, the current interpretation, the completed trade, and already observable outcomes of earlier trades. It is updated after trading and used from the next decision. Here, "short-term" and "long-term" distinguish how information accumulates, not forecast horizons. Appendix C specifies the inputs, exclusions, initialization, and update timing Table 13, the six topics covered Table 14, and translated output examples.

3.3 Community Interaction

The simulated community represents an online stock-discussion forum. The 70 agents with D1 or D2 access can each write one optional post, with a body of at most 500 characters, and read up to five other agents' posts. They select from the same fixed list of authors, titles, post types, and reaction scores, then may like or dislike what they read. Reactions therefore cannot change the list during selection.

The next morning, the five posts with the highest likes minus dislikes are shared with all agents, excluding each author's own posts without replacement. D1 and D2 agents also retain their selected posts; D2 agents receive additional information about authors' holdings and trades. D0 agents receive only the top-post broadcast and cannot post or select posts. Appendix D describes these participation rights and the information delivered to each group.

No community posts are supplied before PM trading, and only claims supported by text the agent has read enter its short-term belief. This shared information can transmit advice effects between agents. Appendix D.2 reports posting and reading activity in this simulation, including how widely posts were read.

Table 3: The two advisors. Lengths count characters in the Korean texts delivered to agents.

	General (A_g)	Personalized (A_p)
Author	Researchers	LLM (GPT 6 Astra)
Recipients	Same text for all 100 agents	One text per agent
Inputs	Common investment principles	Structured profile attributes, latest long-term belief, 90 PRE trades with rationales, trading statistics, three pattern scores
Length	302	At most 500 (mean 423)
Focus	The three patterns and risk management in general terms	The agent's own recurring behavior
Example	"Do not buy more simply because the price has fallen."	"Mandatory trading does not require buying the maximum quantity." (a contrarian agent)

3.4 Advice Generation and Delivery

We compare two fixed advice texts addressing the same three behavioral patterns Table 3. General advice is researcher-written and shared by all agents; personalized advice is LLM-generated separately for each agent from its structured profile, latest long-term belief, all 90 PRE trades with rationales, trading statistics, and pattern scores (Figure 1(b)). Both are prepared once using information available at the end of PRE, without dialogue, and inserted from the next decision into market-analysis, trading-decision, and long-term-belief prompts, but not short-term beliefs or community posts. Appendix G details delivery and reproduces the general and case-study advice.

The two texts share the same heading but also differ in authorship, length, and specificity, so their contrast does not isolate personalization; we treat it as a contrast between bias-specific, agent-level advice and generic advice. Both texts name the patterns we measure (Section 4.3), and some personalized instructions mirror the measures directly. Agent A001, a fundamental-analysis agent positive for all three patterns in PRE, was told to apply the same selling criteria to gains and losses. A lower score may therefore reflect compliance rather than a corrected bias. Conversely, the general text advises securing some gains, which may raise gain-state selling, so A_g serves as a reference for generic advice rather than a neutral baseline.

4 Experimental Setup

We compare no advice, general advice, and personalized advice over 90 trading days in Samsung Electronics. After 45 days without advice, identical copies of each agent's state enter three 45-day branches with common prices and news.

4.1 Data and Execution Settings

Agents trade Samsung Electronics (005930) and hold cash over 90 trading days, from February 27 to July 9, 2026, for 180 decisions per

agent in total. Using one stock allows direct comparison with stock-level retail trading data from the Korea Exchange (KRX). Table 4 lists the initial cash distribution, trading rules, and model.

Table 4: Trading environment. KRW denotes Korean won.

Component	Setting
Assets	Samsung Electronics (005930) and cash
Horizon	90 trading days; 180 decisions per agent
Initial cash	KRW 100 million (73 agents); KRW 1 billion (27 agents)
Actions	Buy or sell; minimum one share; no hold action
Purchase limit	50% of pre-trade available cash
Sale limit	Pre-trade share holdings
Execution	Full fill at AM open or PM close; no price impact
Model	qwen/qwen3.5-flash-02-23

All agents start with cash and no shares. They must buy or sell at least one share at every decision, with purchases limited to half their available cash and sales limited to their holdings. Trades incur no commission or transaction tax. These constraints simplify execution but exclude the option to wait without trading.

News is drawn from a fixed historical collection, targeting five Samsung Electronics articles, three semiconductor-industry articles, and two macroeconomic articles per decision. Missing articles are not replaced with unrelated or future news.

4.2 Shared Period and Advice Conditions

The first 45 trading days, February 27–May 4, form a shared period without advice (PRE). We then copy every agent's profile, views, portfolio, trading history, and community state (posts and reactions) into three versions of the simulation: no advice (A^-), general advice (A_g), and personalized advice (A_p), whose texts are described in Section 3.4 and Table 3. Both advice texts are prepared once from information available by May 4 and delivered from May 6. Each version continues for 45 trading days, May 6–July 9, with the same prices, news, rules, LLM, and random seed for text generation, so each agent can be compared with itself across conditions. Agents' portfolios and community posts can develop differently after advice begins. Each branch was run once. Figure 1(c) shows the design.

4.3 Evaluation Metrics and Comparisons

Alignment with retail trading. We assess whether the agents reproduce the daily buying and selling direction of actual Korean retail investors. We use the full 90-day simulation without advice. For each day, we sum purchase value minus sale value across all agents and both sessions, and compare this *net purchase value* with KRX retail-investor data. We also calculate net share volume. We omit the first three days to reduce the effect of starting entirely in cash, leaving 87 evaluation days. This exclusion applies only to this alignment check.

We measure direction agreement, balanced accuracy (the average agreement rate on actual net-buy and net-sell days), and Pearson correlation of daily net purchases. Four simple rules provide directional benchmarks on the same dates: always net buy; follow the previous day's close-to-close stock return; take the opposite direction; or follow the previous day's retail trading direction. Section 5.1 reports the alignment results and pre-advice behavior.

Behavioral measures. We measure three patterns documented among retail investors [1, 8, 9, 15, 20, 22]: the disposition effect, averaging down, and buying after price surges, each adapted to a single stock with mandatory trading. For agent i at decision t, let P_t be the execution price, $C_{i,t}$ the average acquisition cost, $K_{i,t}$ the available cash, $H_{i,t}$ the shares held, and $q_{i,t}$ the executed quantity. A position is at a gain if $P_t > C_{i,t}$ and at a loss if $P_t < C_{i,t}$. We use only *eligible opportunities*, at which both buying and selling are feasible:

$$H_{i,t} \geq 1 \quad \text{and} \quad Q_{i,t}^{\max} = \left\lfloor \frac{0.5\,K_{i,t}}{P_t} \right\rfloor \geq 1, \tag{1}$$

where 0.5 is the per-trade purchase limit.

The *disposition effect* is the tendency to sell winners more readily than losers [15, 20]. With PSW_i and PSL_i the shares of eligible gain- and loss-state opportunities that ended in a sale,

$$\mathrm{DE}_i = \mathrm{PSW}_i - \mathrm{PSL}_i. \tag{2}$$

Averaging down is buying more of a losing position [21, 22]. Because buying and selling are the only actions, how often an agent buys at a loss mirrors DE, so we measure how much it buys. For eligible purchases with $Q_{i,t}^{\max} \geq 2$, let $u_{i,t} = (q_{i,t} - 1)/(Q_{i,t}^{\max} - 1)$ be the share of additional feasible shares bought beyond the mandatory one. With $\bar{u}_{i,\ell}$ and $\bar{u}_{i,g}$ its means at a loss and at a gain,

$$\mathrm{AD}_i^{\mathrm{size}} = \bar{u}_{i,\ell} - \bar{u}_{i,g}. \tag{3}$$

For *buying after price surges* [1, 8], day d is a surge-response day if the previous day's return was positive and above the 95th percentile of the preceding 252 daily returns. Using the value-based buy–sell imbalance $\mathrm{VBSI} = (\sum B - \sum S)/(\sum B + \sum S)$,

$$S_i^{\mathrm{excess}} = \mathrm{VBSI}_i^{\mathrm{surge}} - \mathrm{VBSI}_i^{\mathrm{non}}. \tag{4}$$

An agent is *positive* for a pattern when both compared states are observed and its score exceeds $\epsilon = 10^{-12}$; for post-surge buying, it must also be a net buyer after surges ($\mathrm{VBSI}_i^{\mathrm{surge}} > \epsilon$). This sign-based classification records an asymmetry in trades, not a statistically significant effect or a psychological cause; DE in particular is not conditioned on recent returns [2].

Comparisons. We compare each agent with itself across branches and use A^-, not PRE, as the reference because PRE covers a different calendar period, so changes from PRE to the post-branch period can arise without any advice. Because each branch was run once and agents interact through the community, the reported p-value is nominal.

5 Experimental Results

We compare advice conditions over the same 45 post-branch days, using PRE only as context. Personalized advice was associated with lower DE than general advice, with less consistent differences in other patterns and investment outcomes.

5.1 Alignment and Pre-Advice Behavior

The simulation broadly reproduced the direction of aggregate retail trading. Over the 87 evaluation days in the full no-advice path, simulated and observed net purchases had the same sign on 70 days (80.46%), exceeding the strongest simple directional benchmark, always net buy (62.07%), by 18.39 percentage points (pp). Balanced

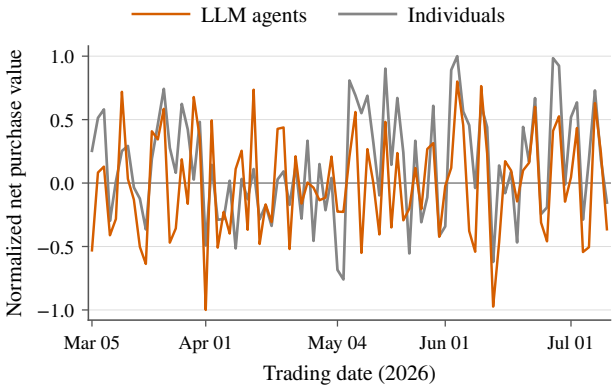


Figure 2: Daily net purchase value of simulated agents and individual investors, each normalized by its own maximum absolute daily value over the 87 evaluation days.

accuracy was 82.49%, and daily net purchases were positively correlated in value ($r = 0.6171$) and share volume ($r = 0.5767$). Table 5 reports the benchmarks, and Figure 2 compares the normalized value series. These results support aggregate alignment, not the validity of individual responses to advice. Because agents observe current prices while the benchmarks use past information, they also do not establish forecasting superiority.

Table 5: Aggregate trading alignment over 87 days in the no-advice path. Agreement measures matching net-buy/net-sell directions; balanced accuracy averages agreement on observed net-buy and net-sell days.

Method	Agreement	Bal. acc.	Pearson's r
LLM (value)	**80.46%**	**82.49%**	0.6171
Always net buy	62.07%	50.00%	–
Lagged return	57.47%	57.49%	–
Contrarian return	42.53%	42.51%	–
Lagged flow	45.98%	42.34%	–

Note. $N = 87$; LLM directions agreed on 70 days. Baseline direction metrics apply to both value and volume because their observed direction labels coincide. Correlations are reported for quantitative flows, not sign-only baselines.

Before advice, the disposition effect was the most widespread measured pattern: 86 of 100 agents sold more often when their position was at a gain than at a loss, with a mean difference of 26.74 pp across all 100 agents. Of 9,000 PRE trades, 6,577 (73.1%) occurred when both buying and selling were feasible. Averaging down—using a larger share of feasible buying capacity when holding a loss—was positive for 60 agents, with a mean intensity score of 4.17 pp across all 100 agents. Post-surge buying was positive for 36 agents, based on six surge-response days. Assigned selling-bias tendency correlated only weakly with PRE disposition-effect scores ($r = 0.213$), and 29 of the 33 agents assigned a low tendency still had a positive score. Persona labels therefore cannot substitute for measuring trades.

5.2 Behavioral Changes Across Advice

The clearest difference across advice conditions was a narrower gap between selling at a gain and selling at a loss. Mean disposition-effect (DE) scores were 36.20, 24.08, and 16.45 pp under no advice, general advice, and personalized advice, respectively Table 6. Relative to no advice, DE was lower for 74 agents with general advice and 87 with personalized advice. Personalized advice also had a 7.63 pp lower mean than general advice, with 64 of 100 agents scoring lower (nominal two-sided exact paired sign test $p = 6.6 \times 10^{-3}$); this is descriptive evidence from one interacting population, not evidence of stability across independent runs.

The decomposition shows what the lower DE represents. Under personalized advice, gain-state and loss-state sale rates were 85.01% and 68.55%, compared with 72.66% and 36.46% without advice. Both rates rose, but the larger increase in loss-state selling narrowed the gap, so the reduction did not come from selling winners less often. Nor did it remove the asymmetry: 80 and 77 of the 86 PRE-positive agents remained positive under general and personalized advice. Because portfolios and eligible opportunities diverge after branching, this describes observed trades rather than a psychological mechanism Appendix H.1.

Averaging-down intensity showed no clear response. Among the 90 agents with defined scores in every post-branch condition, its mean was 2.88, 3.00, and 2.71 pp under A^-, A_g, and A_p. The small differences provide no clear evidence of a reduction in this run.

Post-surge buying was difficult to attribute to advice because it nearly disappeared even without advice: across the nine post-branch surge-response days, only 5, 1, and 1 agents were positive under A^-, A_g, and A_p, compared with 36 in PRE. A before–after comparison alone would thus be misleading, and with few events and a single market path, no incremental advice response can be established Appendix H.5.

5.3 Investor Differences and Illustrative Cases

The additional DE reduction under personalized advice increased from 0.38 pp in the low selling-bias-tendency group to 9.09 pp in the medium group and 13.36 pp in the high group Table 8. Among 19 PRE-positive agents whose DE did not fall under general advice, 14 had lower DE under personalized advice than under no advice.

These group differences are exploratory. High-tendency agents had more room for DE to decline and were disproportionately assigned a technical-analysis strategy. The same gradient was not consistent within the fundamental-analysis group Appendix H.3. The results therefore do not show that more biased investors generally benefit more from personalization; the subgroup selected for a weak response to general advice may also be affected by regression to the mean.

Table 7 contrasts A026 and A093. On June 4 PM, A026 cited advice against buying solely after a decline and sold all 96 shares under personalized advice, versus buying 10 under general advice. Because holdings had already diverged, this contrast does not identify the effect of a specific sentence. A093 cited advice yet sold at only 27.8% of eligible loss-state opportunities and had higher DE than without advice. Advice references therefore do not establish compliance, and their absence does not rule out influence Appendix H.4.

Table 6: Selling behavior and averaging down before and after advice. Gain/loss sale rates are the shares of eligible opportunities at which agents sold when prices were above/below their average acquisition cost; their difference is the disposition effect (DE). Post-branch conditions share calendar dates and initial investor states.

	Disposition effect			Agents (of 100)		Averaging down	
Condition	Gain sale	Loss sale	DE (pp)	DE positive	DE lower than A^-	Positive/defined	Common-90 (pp)
PRE	64.85%	38.12%	26.74	86	–	60/100	–
No advice (A^-)	72.66%	36.46%	36.20	91	–	58/100	2.88
General (A_g)	82.21%	58.13%	24.08	84	74	59/99	3.00
Personalized (A_p)	85.01%	68.55%	16.45	80	87	51/91	2.71

Note. Sale rates and DE are unweighted means across all 100 agents. Common-90 averages averaging-down intensity only for the 90 agents with defined scores in all three post-branch conditions. PRE covers different dates and is shown for context. A positive score denotes a measured pattern, not statistical significance.

Table 7: Contrasting selected responses to advice. A026 shows a large DE reduction; A093 shows why mentioning advice is insufficient evidence of behavioral improvement. These cases are illustrative, not representative of population frequency.

Agent	Advice focus	Observed trading	DE (pp) $A^-/A_g/A_p$	Interpretation
A026: lower DE	Reconsider repeated large purchases at a loss using the reason for holding and future risk.	June 4 PM: bought 10 shares under general advice; sold its 96-share holding under personalized advice.	27.14 / 41.88 / 2.31	A lower DE did not extend to averaging down: that score rose from −39.95 to 9.15 pp (A_g to A_p).
A093: higher DE despite advice references	Use selling criteria independent of gains/losses and distinguish account returns, position returns, and stock allocation.	Under personalized advice, sold at only 27.8% of eligible loss-state opportunities.	12.34 / 14.76 / 51.53	Advice appeared in beliefs and rationales, but executed trades retained a larger gain/loss selling asymmetry.

Table 8: Mean DE (pp) by assigned selling-bias tendency. The last column shows the additional reduction under personalized rather than general advice. Groups describe persona instructions, not measured bias severity.

Tendency (n)	A^-	A_g	A_p	$A_g - A_p$
Low (33)	29.97	17.21	16.83	0.38
Medium (34)	36.44	20.90	11.81	9.09
High (33)	42.18	34.22	20.86	13.36

Table 9: Portfolio outcomes over the common 45-day post-branch period. Entries are unweighted means across 100 agents.

Condition	Return	Max. drawdown	Cash share
No advice	5.83%	12.43%	72.50%
General	3.14%	8.90%	83.37%
Personalized	2.62%	7.37%	87.18%

Note. Returns use each agent's copied May 4 closing wealth. Drawdowns use initial and daily closing account values; cash shares average post-trade cash divided by account value. No commissions or transaction taxes are charged.

5.4 Investment Outcomes and Interpretation

Both advice conditions had higher cash shares, lower returns, and smaller drawdowns than no advice Table 9, along with lower turnover Appendix H.2. Lower DE therefore accompanied reduced stock exposure and a different return–risk outcome; it did not establish better investment outcomes or benefits for real investors.

6 Conclusion

We developed an LLM-agent simulation that compares advice designs from matched investor states under fixed historical prices and news. In this exploratory run, personalized advice was associated with a weaker disposition effect than general advice, driven by a larger increase in loss-state selling rather than a decrease in gain-state selling; the asymmetry narrowed but did not disappear for most agents. The additional reduction was most apparent where general advice had changed little, yet responses varied across investors, and citing the advice in stated reasoning did not necessarily change trades. Because lower behavioral scores accompanied higher cash shares, lower returns, and smaller drawdowns, advice should be evaluated through executed trades across several behavioral measures and performance outcomes jointly, rather than through average scores or stated acceptance alone.

As an exploratory study, this work identifies patterns worth testing rather than establishing general effects. Each branch is a single trajectory of interacting agents rather than an independent replication, the two advice texts differ in authorship, length, and specificity, and aggregate alignment with retail trading does not validate individual responses to advice; the single-stock setting, mandatory trading, assigned tendencies, and fixed price path further limit generalization to real investors. Because the testbed holds investor states and market paths fixed, it can be extended directly to address these limits: repeating the comparison across seeds, stocks, and periods, adding controls that separate investor-specific information from other differences in the advice texts, and generating hypotheses about advice design that can then be tested with real investors.

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Appendix

A Data and Environment

Prices. Prices are Samsung Electronics (005930.KS) OHLCV from Yahoo Finance (`yfinance`, `auto_adjust=False`). Execution uses the unadjusted open in AM and the unadjusted close in PM, not the adjusted-close series; cash dividends and corporate actions are not credited by the execution ledger. The path is a fixed retrospective series, and PM decisions occur after the close is known. Figure 3 marks the branch boundary and the nine post-branch surge-response days.

News. The sealed news bundle extends through July 22; only its February 27–July 9 event slots are used. The 180 event boards contain 1,474 articles (all distinct) out of 1,800 target slots (18.1% missing); a shortage in one category is filled neither from another category nor from future news. Shortages occur in 59/90 PRE events (65.6%; 760 articles) and 69/90 post-branch events (76.7%; 714 articles), 128/180 (71.1%) in total. Source fields label 1,442 articles as `mk` (Maeil Business Newspaper) and 32 as `repository-split`, without original-publisher metadata. Agents see titles and summaries according to news depth; raw article bodies are never delivered, and body hashes are retained for provenance.

Retail-investor flow. Retail flow is the KRX Information Data System's daily net-purchase series by investor type for Samsung Electronics, Individuals category, in purchase value (KRW) and share volume (source coverage February 27–July 22). Positive values are net-buy days and negative values net-sell days; no evaluation date has zero flow. Simulated flow is the signed sum of AM and PM fills, with purchases positive.

Model. Trading uses `qwen/qwen3.5-flash-02-23` through OpenRouter's Alibaba provider (fallback disabled, reasoning effort none). Each agent-event makes four core calls (STB, analysis, decision, post-fill LTB); search and community stages add calls. Initial temperatures are 0.2 for the core stages, search, and reactions, 0.3 for selection and community interpretation, and 0.7 for posting. Schema-invalid responses allow up to ten attempts, with retry temperatures rising toward 1.0 without relaxing validation; transport retries are separate.

B Persona Construction

Persona prompts begin with the Korean equivalent of "You are a Korean retail investor in Samsung Electronics" and list user type, selling-bias tendency, lottery preference, past performance, diversification preference, analysis strategy, entry timing, news depth, and initial cash, in that order; gender, age, and region are omitted. Table 10 summarizes selected instructions (summaries, not verbatim stimuli). Selling-bias tendency corresponds to the disposition-tendency attribute of the source profiles and also includes instructions about buying after losses. Entry timing applies only to new purchases, whereas analysis strategy specifies valuation evidence. Attributes are not independently distributed (Tables 11 and 12): high selling-bias tendency occurs in 28/60 technical but only 5/40 fundamental personas, so subgroup results do not identify an isolated selling-bias-tendency effect. All nine influencers have D1 or D2 access.

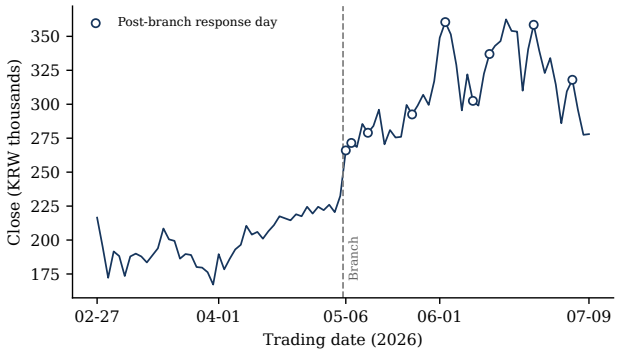


Figure 3: Fixed Samsung Electronics close-price path. The dashed line marks the boundary between May 4 (last PRE day) and May 6 (first branch day). Circles mark the nine post-branch surge-response days rather than the preceding signal days.

C Belief Representations

STB and LTB share the six dimensions in Table 14, each limited to 150 characters; Table 13 gives their inputs and timing. The following English translations are A022's Korean outputs at April 10 PM. They are model judgments, not assessments endorsed by the authors.

STB reflection. "Information limits: no directly relevant Samsung Electronics news or numerical data. Caution: do not infer that the stock will rise solely from related-stock gains; fundamental analysis is needed."

LTB reflection. "The recent t050 purchase was successful with a 10.6% rise, whereas the t059 purchase fell by 1.2%, indicating a timing error. Reduce speculative purchases under limited information and focus on fundamental analysis."

D Community

D.1 Selection and Ranking Details

After PM execution and the post-fill LTB, each D1/D2 agent may post once and then selects from a frozen board that excludes its own posts. The selection screen shows pseudonym, title, type, and reaction counts and score, but no body or author portfolio. Selecting zero posts is valid, and only selected bodies are read and reacted to (like, dislike, or none). The global Best five are ranked by likes minus dislikes, then likes, then ascending post ID, with no positive-score filter and no refill for recipients whose own post is excluded. The next AM delivers the Best and selected bodies, serializing a body shared by both routes once. D2 agents also receive the author's frozen PM portfolio and up to three latest trades. Reputation badges are not provided.

D.2 Activity in PRE

All 70 eligible agents posted on each of the 45 PRE days (3,150 posts) although posting was optional. Mean body length was 338.49 characters, and readers selected 4.985 posts per eligible agent-day. Of 15,702 direct reads/reactions, 822/3,150 posts (26.1%) received at least one direct read. Post types were trade sharing (78.8%),

Table 10: Summaries of selected persona instructions; each level maps to one fixed sentence in the Korean prompt.

Attribute	Level	Instruction summary
Selling bias	High	Sell immediately after gains above 10%; continue buying as prices fall, disregarding risk.
	Medium	Consider selling after a 10% gain and liquidate after 20%; after declines, consider averaging down or cutting losses.
	Low	Look for an exit after a 20% gain; after a decline above 10%, assess cutting losses rationally.
Lottery preference	High	Seek recently surged risky assets, hoping for large overnight gains.
	Medium	Occasionally allocate part of capital to recently surged assets.
	Low	Such assets have little influence on decisions.
Analysis strategy	Technical	Trade quickly using short-term moves, indicators, and trend forecasts.
	Fundamental	Focus on intrinsic value and growth; buy undervaluation and sell overvaluation; pay little attention to technical indicators.
Entry timing	Contrarian	Prefer recent declines for new purchases and avoid chasing sharp rises.
	Neutral	First assess firm value and available information, then price trends.
	Momentum	Prefer steady recent rises for new purchases; avoid buying during declines.
News depth	D0	Read article titles only.
	D1	Read titles and summaries.
	D2	D1 plus searches for up to five articles from the preceding seven days.

Table 11: Joint cohort composition by strategy; H/M/L denote high/medium/low selling-bias tendency and C/N/M contrarian/neutral/momentum timing.

Strategy	N	H	M	L	C	N	M	D0	D1	D2
Technical	60	28	19	13	31	16	13	20	32	8
Fundamental	40	5	15	20	19	19	2	10	23	7

Table 12: Initial cash (KRW) and influencers by news depth.

Depth	N	100M	1B	Small	Large
D0	30	22	8	0	0
D1	55	40	15	6	1
D2	15	11	4	2	0

profit/loss sharing (13.2%), analysis (7.0%), and impressions (1.0%); no questions or columns were generated. Of the 225 Best selections (99 recipients each), 220 had a subsequent AM within PRE (21,780 delivery relations); the five May 4 PM posts reach agents on May 6.

Limit on identifying peer transmission. In the advice branches every agent receives the advice directly, so peer transmission through community posts cannot be separated from direct receipt in this design.

E Alignment with Retail Flow

The initial three days are excluded only for this alignment check, giving 42 PRE and 45 post-branch days. The no-advice results were independently recomputed from signed fills and the raw KRX series. Table 15 splits the no-advice comparison by period (34 + 36 = 70 matches, as in the main text) and adds the advice branches over the same 45 post-branch dates. Lower alignment under advice does not by itself invalidate a behavioral intervention, since matching observed retail flow is not a welfare target.

F Measurement Opportunities

PRE contains 9,000 executions (5,755 purchases, 3,245 sales). The 2,423 single-direction executions are excluded from behavioral measures, leaving the 6,577 eligible opportunities reported in the main text; the three-day alignment exclusion is not applied to behavioral scores. Table 16 reports agent-level denominators: gain/loss opportunities for DE and eligible purchases for averaging-down size, which additionally require at least two feasible shares. Undefined scores are neither recoded as zero nor counted as mitigation.

G Advice Delivery and Stimuli

G.1 Delivery

From May 6, the assigned body is inserted into the market-analysis, trading-decision, and post-fill LTB prompts; it is not inserted into STB or community posting. Both advice branches share a common heading, which translates as "A memo from your assigned investment advisor:" (the Korean original is omitted for font compatibility). The following schematic is not a verbatim prompt:

> Analysis: persona → prior LTB/current STB → market and account state → advice → output instructions.
> Decision: persona → beliefs → analysis → portfolio/order history → constraints → advice → decision instructions.
> LTB: structured prior LTB/current STB, fill, matured outcomes, and advice → update contract.

Agents receive only the advice body (1–500 Korean characters, enforced without truncation), not the evidence generated with it. The quotations below translate the delivered Korean stimuli; character counts refer to the Korean originals, and normative wording is retained.

G.2 General Advice

> Before trading, calmly review the available information and your holdings. Even when the outlook looks favorable, check whether the expectation is sufficiently supported before increasing your investment. Do not buy more simply because the price has fallen or rush to recover losses; watch whether uncertainty is declining. When gains arise, secure some of

Table 13: Inputs and timing of the two belief representations.

Component	STB	LTB
Generated	Every event, before decision	Every event, after actual execution
Inputs	Persona, current news, community bodies actually read	Previous LTB, current STB, current execution, matured trade outcomes; advice if assigned
Not shown	Prior beliefs, prices and indicators, account state, earlier trades	Future outcomes
Used in	Current analysis and decision	Next-event analysis and decision; PM community posting
Evidence	Supporting/opposing news and post IDs per dimension	Inherited STB evidence; outcome IDs only in self-reflection
Initialization	None	Rule-based strategy-specific neutral sentences; no LLM call

Table 14: Six belief dimensions shared by STB and LTB.

Dim.	Content
1	Price direction over approximately the next month
2	Valuation and supporting reasoning
3	Rates, exchange rates, economy, semiconductor conditions
4	Market sentiment and investor mood
5	Persona-specific news and community interpretation
6	Judgment and risk reflection; LTB also reviews fills and outcomes

Table 15: Direction agreement with KRX individual-investor flow and value-flow correlation.

Path (period)	Match	Agree.	Bal. acc.	Always buy	r_{value}
A^-, PRE (42 d)	34/42	80.95%	83.06%	59.52%	–
A^-, post-branch (45 d)	36/45	80.00%	81.68%	64.44%	0.7076
A_{g}, post-branch (45 d)	31/45	68.89%	70.26%	64.44%	0.5097
A_{p}, post-branch (45 d)	32/45	71.11%	71.98%	64.44%	0.5796

Note. Balanced accuracy averages recalls for observed net-buy and net-sell days. Post-branch volume correlations are 0.6981 (A^-), 0.4884 (A_{g}), and 0.5776 (A_{p}); value and volume give identical direction metrics.

them to reduce exposure rather than pursuing further increases, and when losses arise, do not postpone judgment solely in the hope of a recovery. After a sharp rise, reassess the price and evidence instead of chasing the stock out of regret over a missed opportunity. Keep cash as flexibility for the next decision, and review the reasons and outcomes after each trade.

G.3 Personalized Advice: Case-Study Agents

A026.

To avoid averaging down to recover your initial investment, do not fill the purchase limit merely because the price has declined; first check whether current information undermines the original purchase rationale. Repeated use of the maximum limit brought your stock weight to 99.4%, which you described as preserving room to respond. Show confidence in concentrated investing through the loss limit you can maintain, rather than the amount invested. To avoid the disposition effect, abandon the rule of immediately securing small gains while waiting for losses to rebound. Hide the purchase price and decide whether to sell based on future risk and the rationale for holding. To avoid chasing the previous day's surge, do not assume that a pullback after a surge is a lottery-like opportunity; consider adverse evidence in the supplied news summaries and further losses together. Distinguish position returns, account returns, price changes, and actual portfolio weight at every decision. Two recent successful rebounds or a rise after you sold do not prove that the strategy is correct.

A093.

Your selling rate was higher in gain states than in loss states. To avoid the disposition effect, use prospective intrinsic value and risk, rather than a profit-taking target, as the basis for selling. You described a cumulative account return of 20% as a 20% rise in the shares held, but the actual position return at the last trade was approximately 2%. Separate account performance from the gains and losses on current holdings, and use neither number as evidence that the current price is expensive. Distinguish a price below average acquisition cost from objective undervaluation. Review changes in earnings and demand alongside labor-cost and substitute-technology risks in the supplied summaries; apply criteria independent of gain or loss when the value outlook is the same. You called staggered buying diversification, although stock weight was nearly 98%. Consider total accumulated holdings and cash rather than only the order amount. Do not treat rebounds and final profits as proof that the rule is correct; revisit the opposing evidence available at the time as well.

G.4 Advice Characteristics

The 100 personalized bodies range from 333 to 491 characters (quartiles 390.75 and 454.5; the 500-character limit is a ceiling). A lexical screen finds disposition-related wording (the term disposition effect or selected selling-winners/holding-losers expressions) in 92/100 texts, averaging-down wording (including lowering acquisition cost or recovering break-even) in 75/100, surge or chasing wording in 49/100, and gain/loss-independent or same-selling-criterion wording in 58/100. These are reproducible string-pattern counts, not validated semantic coding; the full regexes and all original texts are provided in the supplementary material. Their overlap with the outcome definitions supports treating the contrast as bias-specific advice rather than personalization alone.

H Additional Results

H.1 Disposition Effect: Gain/Loss-State Decomposition

Table 17 uses unweighted agent means, so mean DE equals mean P_{SW} minus mean P_{SL}. Relative to A^-, gain-state selling rises by 9.55

Table 16: Agent-level measurement denominators (minimum–median–maximum).

Arm	DE gain	DE loss	AD gain buys	AD loss buys	AD undefined
PRE	18–31–49	19–35–60	4–11–23	3–22–44	0
A^-	20–29–39	19–34–47	2–8–18	1–23–40	0
A_g	23–29–48	14–31–46	1–5–19	0–13–30	1
A_p	20–25.5–35	19–26.5–42	0–4–16	0–7–27	9

Note. All DE scores are defined. AD has 100, 100, 99, and 91 defined scores in PRE, A^-, A_g, and A_p; 90 agents have defined AD in all three post-branch conditions.

Table 17: Disposition-effect decomposition (agent means).

Arm	P_{SW}	P_{SL}	DE (pp)
PRE	64.85%	38.12%	26.74
A^-	72.66%	36.46%	36.20
A_g	82.21%	58.13%	24.08
A_p	85.01%	68.55%	16.45

Note. All 100 agents have both gain and loss opportunities in each condition.

Table 18: Descriptive performance (unweighted agent means).

Arm	Return	MDD	Cash share	Turnover
PRE	9.25%	11.53%	68.22%	19.52
A^-	5.83%	12.43%	72.50%	20.00
A_g	3.14%	8.90%	83.37%	13.32
A_p	2.62%	7.37%	87.18%	14.01

Note. PRE and post-branch periods cover different market paths.

pp under A_g and 12.34 pp under A_p, while loss-state selling rises by 21.67 and 32.09 pp: the average reduction reflects a larger increase in loss-state selling, not a decrease in gain-state selling. Among the 74 agents with lower DE under A_g, 51 have higher loss-state selling without lower gain-state selling, 23 have both changes, and none have only lower gain-state selling; among the 87 A_p reducers the counts are 70, 16, and 1. These categories describe directions, not causal contributions, and gain/loss states and opportunity denominators themselves change across branches.

H.2 Descriptive Performance Statistics

Table 18 reports descriptive performance. Post-branch returns use each agent's copied May 4 PM wealth as the denominator (PRE returns use initial cash); MDD uses the initial endpoint and daily PM marked wealth; cash share is the mean cash/equity ratio over post-fill states; turnover is gross purchase plus sale value divided by the period's initial equity (not annualized). Advice accompanies lower returns, higher cash shares, and lower turnover on the same market path, so a lower DE alone does not establish improved outcomes.

H.3 Strategy and Selling-Bias Tendency

High selling-bias tendency is concentrated among technical personas (Table 11), so Table 19 crosses strategy with assigned tendency. The $A_p - A_g$ contrast deepens across technical low/medium/high groups but is not monotonic among fundamental personas (whose high cell has five agents), so the pooled gradient should not be read as a strategy-independent personalization effect.

Table 19: Joint strategy and selling-bias-tendency splits of DE (pp).

Strategy	Selling bias	N	A^-	$A_g - A^-$	$A_p - A^-$	$A_p - A_g$
Technical	Low	13	20.61	−7.97	−14.61	−6.64
Technical	Medium	19	30.58	−14.99	−25.99	−11.00
Technical	High	28	40.57	−6.88	−22.49	−15.62
Fundamental	Low	20	36.05	−15.88	−12.19	+3.68
Fundamental	Medium	15	43.86	−16.22	−22.92	−6.69
Fundamental	High	5	51.22	−14.06	−14.72	−0.66

Note. Descriptive cell means; sparse, correlated groups do not identify independent effects.

H.4 Explicit Advice Wording in Rationales and LTB

We calculated text-based proxies for advice compliance rates using a lexical audit of agents' trading rationales and long-term beliefs. The advice combines qualitative and conditional rules and does not prescribe a unique trade at every event. Screening decision rationales (including risk-control text) and all six LTB dimensions for explicit advisor/advice/consultation wording flags rationales in 0, 7, 171 (1.9%), and 4,426 (49.2%) of the 9,000 fills in PRE, A^-, A_g, and A_p, and LTBs in 21 (0.2%), 66 (0.7%), 452 (5.0%), and 6,043 (67.1%). At least one rationale marker appears for 0, 7, 60, and 100 agents, respectively. Baseline matches show that a marker need not refer to the assigned stimulus, the absence of a marker does not establish the absence of influence, and mentioning advice does not establish behavioral compliance (A093 illustrates why). A semantic compliance percentage would require a coding rubric and adjudication of conditional rules and trade context, and is not inferred from these counts.

H.5 Post-Surge Buying: Coverage and Sensitivity

The nine post-branch response dates are May 6, May 7, May 12, May 22, June 2, June 10, June 15, June 25, and July 6; AM and PM on a response day share the same lagged signal. Eligible response-day opportunities per agent range over 6–11 in PRE, 9–14 under A^-, 9–16 under A_g, and 9–14 under A_p. Under alternative signal definitions, the number of response days (PRE/post) is 7/12 for the 90th percentile of the 252-day window and 5/5 for the 95th percentile of a 126- or 63-day window; only event coverage was tested, and behavioral effects under alternative definitions were not estimated.

Classification counts are unchanged for $\epsilon = 0, 10^{-12}, 10^{-6}$, and 10^{-3}.

With nine response days and a single fixed market path, the pattern is reported descriptively rather than as an incremental advice response.